5. Address Space

5.1 Address Space

The MCU supports a 4 GB linear address space ranging from 0x0000_0000 to 0xFFFF_FFFF that can contain both program and data. In the address from 0x0000_0000 to 0x5FFF_FFFF, secure and non-secure region are isolated. Figure 5.1 and Figure 5.2 show the memory map example of 1 MB MRAM product.

Address	CPU0 (CM85)	CPU1 (CM33)	the others	IDAU/MSAU Security_Attribution
0xFFFF_FFFF	System for CM85	System for CM33	Reserved area*1	
0xE010_0000	Private Peripheral bus	Private Peripheral bus	Reserved area*1	
0xE000_0000	Reserved area*1			Non-secure
0xA000_0000	External address space (OSPI area)			
0x7000_0000	External address space (SDRAM area)			
0x6800_0000	External address space (CS area)			
0x6000_0000	Reserved area*1			
0x5050_0000	Peripheral IO registers			
0x5000_0000	Reserved area*1			Secure
0x4050_0000	Peripheral IO registers			
0x4000_0000	Reserved area*1			Non-secure
0x3A02_0000	CPU1-STCM alias*9	Reserved area*1	CPU1-STCM alias*9	
0x3A01_0000	CPU1-CTCM alias*8	Reserved area*1	CPU1-CTCM alias*8	
0x3A00_0000	Reserved area*1			
0x3804_0000	Reserved area*1	CPU0-DTCM alias*7		
0x3802_0000	Reserved area*1	CPU0- ITCM alias*6		
0x3800_0000	Reserved area*1			
0x321D_4000	On-chip SRAM (SRAME)*10			
0x321A_0000	On-chip SRAM			
0x3200_0000	Reserved area*1			
0x3002_0000	CPU0-DTCM*7	Reserved area*1	Reserved area*1	
0x3001_0000		CPU1-STCM*9		
0x3000_0000	Reserved area*1			Non-secure callable for CPU Secure for other bus masters
0x2A02_0000	CPU1-STCM alias*5	Reserved area*1	CPU1-STCM alias*5	
0x2A01_0000	CPU1-CTCM alias*4	Reserved area*1	CPU1-CTCM alias*4	
0x2A00_0000	Reserved area*1			
0x2804_0000	Reserved area*1	CPU0-DTCM alias*3		
0x2802_0000	Reserved area*1	CPU0-ITCM alias*2		
0x2800_0000	Reserved area*1			
0x221D_4000	On-chip SRAM (SRAME)*10			
0x221A_0000	On-chip SRAM			
0x2200_0000	Reserved area*1			
0x2002_0000	CPU0-DTCM*3	Reserved area*1	Reserved area*1	
0x2001_0000		CPU1-STCM*5		
0x2000_0000	Reserved area*1			Non-secure
0x1880_0000	SiP-Flash area (OSPI1 CS1 area)			
0x1800_0000	Reserved area*1			
0x1300_0000	Extra MRAM (option-setting memory)			
0x12E0_0000	Reserved area*1			
0x12D0_0000	Extra MRAM (option-setting memory)			
0x12A0_0000	Reserved area*1			
0x1210_0000	Code MRAM			
0x1200_0000	Reserved area*1			
0x1002_0000	CPU0-ITCM*6	Reserved area*1	Reserved area*1	
0x1001_0000		CPU1-CTCM*8		
0x1000_0000	Reserved area*1			Non-secure callable for CPU Secure for other bus masters
0x0880_0000	SiP-Flash area (OSPI1 CS1 area)			
0x0800_0000	Reserved area*1			
0x0300_0000	Extra MRAM (option-setting memory)			
0x02E0_0000	Reserved area*1			
0x02D0_0000	Extra MRAM (option-setting memory)			
0x02A0_0000	Reserved area*1			
0x0210_0000	Code MRAM			
0x0200_0000	Reserved area*1			
0x0002_0000	CPU0- ITCM*2	Reserved area*1	Reserved area*1	
0x0001_0000		CPU1-CTCM*4		
0x0000_0000				

Note: See Table 5.1. The capacity of the Code MRAM, On-chip SRAM differs depending on the product.
Note 1. Reserved areas should not be accessed.
Note 2. CPU0-ITCM for Secure has different addresses for CPU1 and other bus masters.
Note 3. CPU0-DTCM for Secure has different addresses for CPU1 and other bus masters.
Note 4. CPU1-CTCM for Secure has different addresses for CPU0 and other bus masters.
Note 5. CPU1-STCM for Secure has different addresses for CPU0 and other bus masters.
Note 6. CPU0-ITCM for Non-Secure has different addresses for CPU1 and other bus masters.
Note 7. CPU0-DTCM for Non-Secure has different addresses for CPU1 and other bus masters.
Note 8. CPU1-CTCM for Non-Secure has different addresses for CPU0 and other bus masters.
Note 9. CPU1-STCM for Non-Secure has different addresses for CPU0 and other bus masters.
Note 10. These areas are accessible only when SRAM ECC function is disabled.

Figure 5.1 1 MB MRAM product for SiP Product

Address	CPU0 (CM85)	CPU1 (CM33)	the others	IDAU/MSAU Security_Attribution
0xE010_0000 – 0xFFFF_FFFF	System for CM85	System for CM33	Reserved area*1	
0xE000_0000 – 0xE010_0000	Private Peripheral bus	Private Peripheral bus	Reserved area*1	
0xA000_0000 – 0xE000_0000	Reserved area*1	Reserved area*1	Reserved area*1	Non-secure
0x7000_0000 – 0xA000_0000	External address space (OSPI area)	External address space (OSPI area)	External address space (OSPI area)	Non-secure
0x6800_0000 – 0x7000_0000	External address space (SDRAM area)	External address space (SDRAM area)	External address space (SDRAM area)	Non-secure
0x6000_0000 – 0x6800_0000	External address space (CS area)	External address space (CS area)	External address space (CS area)	Non-secure
0x5050_0000 – 0x6000_0000	Reserved area*1	Reserved area*1	Reserved area*1	Non-secure
0x5000_0000 – 0x5050_0000	Peripheral IO registers	Peripheral IO registers	Peripheral IO registers	Non-secure
0x4050_0000 – 0x5000_0000	Reserved area*1	Reserved area*1	Reserved area*1	Secure
0x4000_0000 – 0x4050_0000	Peripheral IO registers	Peripheral IO registers	Peripheral IO registers	Secure
0x3A02_0000 – 0x4000_0000	Reserved area*1	Reserved area*1	Reserved area*1	Non-secure
0x3A01_0000 – 0x3A02_0000	CPU1-STCM alias*9	Reserved area*1	CPU1-STCM alias*9	Non-secure
0x3A00_0000 – 0x3A01_0000	CPU1-CTCM alias*8	Reserved area*1	CPU1-CTCM alias*8	Non-secure
0x3804_0000 – 0x3A00_0000	Reserved area*1	Reserved area*1	Reserved area*1	Non-secure
0x3802_0000 – 0x3804_0000	Reserved area*1	CPU0-DTCM alias*7	CPU0-DTCM alias*7	Non-secure
0x3800_0000 – 0x3802_0000	Reserved area*1	CPU0- ITCM alias*6	CPU0- ITCM alias*6	Non-secure
0x321D_4000 – 0x3800_0000	Reserved area*1	Reserved area*1	Reserved area*1	Non-secure
0x321A_0000 – 0x321D_4000	On-chip SRAM (SRAME)*10	On-chip SRAM (SRAME)*10	On-chip SRAM (SRAME)*10	Non-secure
0x3200_0000 – 0x321A_0000	On-chip SRAM	On-chip SRAM	On-chip SRAM	Non-secure
0x3002_0000 – 0x3200_0000	Reserved area*1	Reserved area*1	Reserved area*1	Non-secure
0x3001_0000 – 0x3002_0000	CPU0-DTCM*7	Reserved area*1	Reserved area*1	Non-secure
0x3000_0000 – 0x3001_0000	CPU0-DTCM*7	CPU1-STCM*9	Reserved area*1	Non-secure
0x2A02_0000 – 0x3000_0000	Reserved area*1	Reserved area*1	Reserved area*1	Non-secure callable for CPU Secure for other bus masters
0x2A01_0000 – 0x2A02_0000	CPU1-STCM alias*5	Reserved area*1	CPU1-STCM alias*5	Non-secure callable for CPU Secure for other bus masters
0x2A00_0000 – 0x2A01_0000	CPU1-CTCM alias*4	Reserved area*1	CPU1-CTCM alias*4	Non-secure callable for CPU Secure for other bus masters
0x2804_0000 – 0x2A00_0000	Reserved area*1	Reserved area*1	Reserved area*1	Non-secure callable for CPU Secure for other bus masters
0x2802_0000 – 0x2804_0000	Reserved area*1	CPU0-DTCM alias*3	CPU0-DTCM alias*3	Non-secure callable for CPU Secure for other bus masters
0x2800_0000 – 0x2802_0000	Reserved area*1	CPU0-ITCM alias*2	CPU0-ITCM alias*2	Non-secure callable for CPU Secure for other bus masters
0x221D_4000 – 0x2800_0000	Reserved area*1	Reserved area*1	Reserved area*1	Non-secure callable for CPU Secure for other bus masters
0x221A_0000 – 0x221D_4000	On-chip SRAM (SRAME)*10	On-chip SRAM (SRAME)*10	On-chip SRAM (SRAME)*10	Non-secure callable for CPU Secure for other bus masters
0x2200_0000 – 0x221A_0000	On-chip SRAM	On-chip SRAM	On-chip SRAM	Non-secure callable for CPU Secure for other bus masters
0x2002_0000 – 0x2200_0000	Reserved area*1	Reserved area*1	Reserved area*1	Non-secure callable for CPU Secure for other bus masters
0x2001_0000 – 0x2002_0000	CPU0-DTCM*3	Reserved area*1	Reserved area*1	Non-secure callable for CPU Secure for other bus masters
0x2000_0000 – 0x2001_0000	CPU0-DTCM*3	CPU1-STCM*5	Reserved area*1	Non-secure callable for CPU Secure for other bus masters
0x1800_0000 – 0x2000_0000	Reserved area*1	Reserved area*1	Reserved area*1	Non-secure
0x1300_0000 – 0x1800_0000	Reserved area*1	Reserved area*1	Reserved area*1	Non-secure
0x12E0_0000 – 0x1300_0000	Extra MRAM (option-setting memory)	Extra MRAM (option-setting memory)	Extra MRAM (option-setting memory)	Non-secure
0x12D0_0000 – 0x12E0_0000	Reserved area*1	Reserved area*1	Reserved area*1	Non-secure
0x12A0_0000 – 0x12D0_0000	Extra MRAM (option-setting memory)	Extra MRAM (option-setting memory)	Extra MRAM (option-setting memory)	Non-secure
0x1210_0000 – 0x12A0_0000	Reserved area*1	Reserved area*1	Reserved area*1	Non-secure
0x1200_0000 – 0x1210_0000	Code MRAM	Code MRAM	Code MRAM	Non-secure
0x1002_0000 – 0x1200_0000	Reserved area*1	Reserved area*1	Reserved area*1	Non-secure
0x1001_0000 – 0x1002_0000	CPU0-ITCM*6	Reserved area*1	Reserved area*1	Non-secure
0x1000_0000 – 0x1001_0000	CPU0-ITCM*6	CPU1-CTCM*8	Reserved area*1	Non-secure
0x0300_0000 – 0x1000_0000	Reserved area*1	Reserved area*1	Reserved area*1	Non-secure callable for CPU Secure for other bus masters
0x02E0_0000 – 0x0300_0000	Extra MRAM (option-setting memory)	Extra MRAM (option-setting memory)	Extra MRAM (option-setting memory)	Non-secure callable for CPU Secure for other bus masters
0x02D0_0000 – 0x02E0_0000	Reserved area*1	Reserved area*1	Reserved area*1	Non-secure callable for CPU Secure for other bus masters
0x02A0_0000 – 0x02D0_0000	Extra MRAM (option-setting memory)	Extra MRAM (option-setting memory)	Extra MRAM (option-setting memory)	Non-secure callable for CPU Secure for other bus masters
0x0210_0000 – 0x02A0_0000	Reserved area*1	Reserved area*1	Reserved area*1	Non-secure callable for CPU Secure for other bus masters
0x0200_0000 – 0x0210_0000	Code MRAM	Code MRAM	Code MRAM	Non-secure callable for CPU Secure for other bus masters
0x0002_0000 – 0x0200_0000	Reserved area*1	Reserved area*1	Reserved area*1	Non-secure callable for CPU Secure for other bus masters
0x0001_0000 – 0x0002_0000	CPU0- ITCM*2	Reserved area*1	Reserved area*1	Non-secure callable for CPU Secure for other bus masters
0x0000_0000 – 0x0001_0000	CPU0- ITCM*2	CPU1-CTCM*4	Reserved area*1	Non-secure callable for CPU Secure for other bus masters

Note: See Table 5.1. The capacity of the Code MRAM, On-chip SRAM differs depending on the product.
Note 1. Reserved areas should not be accessed.
Note 2. CPU0-ITCM for Secure has different addresses for CPU1 and other bus masters.
Note 3. CPU0-DTCM for Secure has different addresses for CPU1 and other bus masters.
Note 4. CPU1-CTCM for Secure has different addresses for CPU0 and other bus masters.
Note 5. CPU1-STCM for Secure has different addresses for CPU0 and other bus masters.
Note 6. CPU0-ITCM for Non-Secure has different addresses for CPU1 and other bus masters.
Note 7. CPU0-DTCM for Non-Secure has different addresses for CPU1 and other bus masters.
Note 8. CPU1-CTCM for Non-Secure has different addresses for CPU0 and other bus masters.
Note 9. CPU1-STCM for Non-Secure has different addresses for CPU0 and other bus masters.
Note 10. These areas are accessible only when SRAM ECC function is disabled.

Figure 5.2 1 MB MRAM product for Standard Product

Table 5.1 Capacity of the Code MRAM, SRAM and SiP-Flash

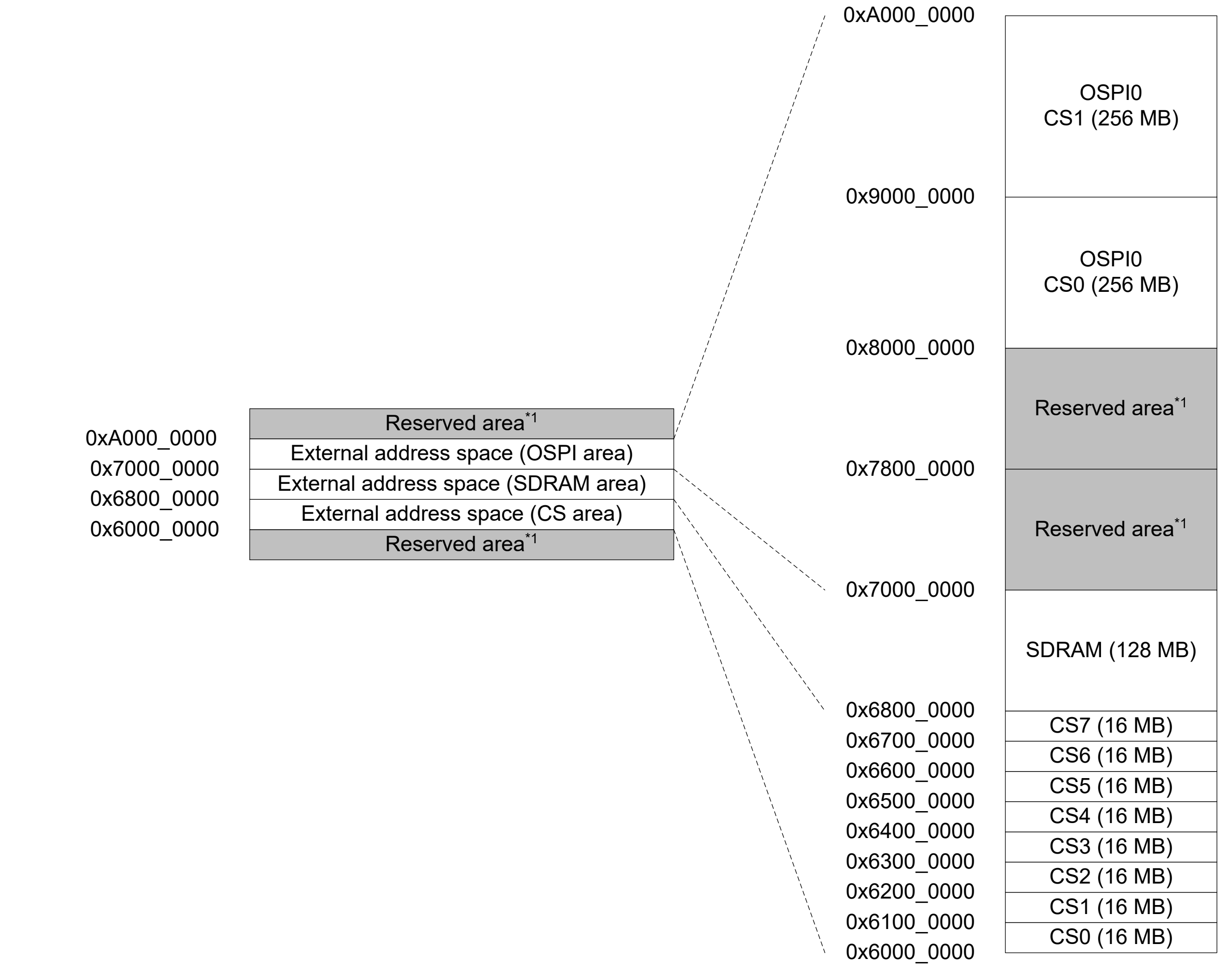
CPU	Code MRAM			On-chip SRAM			SiP-Flash		
	Capacity	Address		Capacity	Address		Capacity	Address	
		Secure alias	Non-Secure alias		Secure alias	Non-Secure alias		Secure alias	Non-Secure alias
Dual	1 MB	0x0200_0000 to 0x0210_0000	0x1200_0000 to 0x1210_0000	1.6 MB	0x2200_0000 to 0x221A_0000	0x3200_0000 to 0x321A_0000	4 MB	0x0800_0000 to 0x0840_0000	0x1800_0000 to 0x1840_0000
							8 MB	0x0800_0000 to 0x0880_0000	0x1800_0000 to 0x1880_0000
Single	1 MB	0x0200_0000 to 0x0210_0000	0x1200_0000 to 0x1210_0000	1.6 MB + 128 KB*1	0x2200_0000 to 0x221A_0000 0x2A00_0000 to 0x2A02_0000	0x3200_0000 to 0x321A_0000 0x3A00_0000 to 0x3A02_0000	—		
	512 KB	0x0200_0000 to 0x0208_0000	0x1200_0000 to 0x1208_0000						

Note 1. In single-core, C-TCM and S-TCM at addresses 0x2A00_0000 to 0x2A01_FFFFF can be used as SRAM. For details of TCM, see section 2, CPU.

5.2 External Address Space

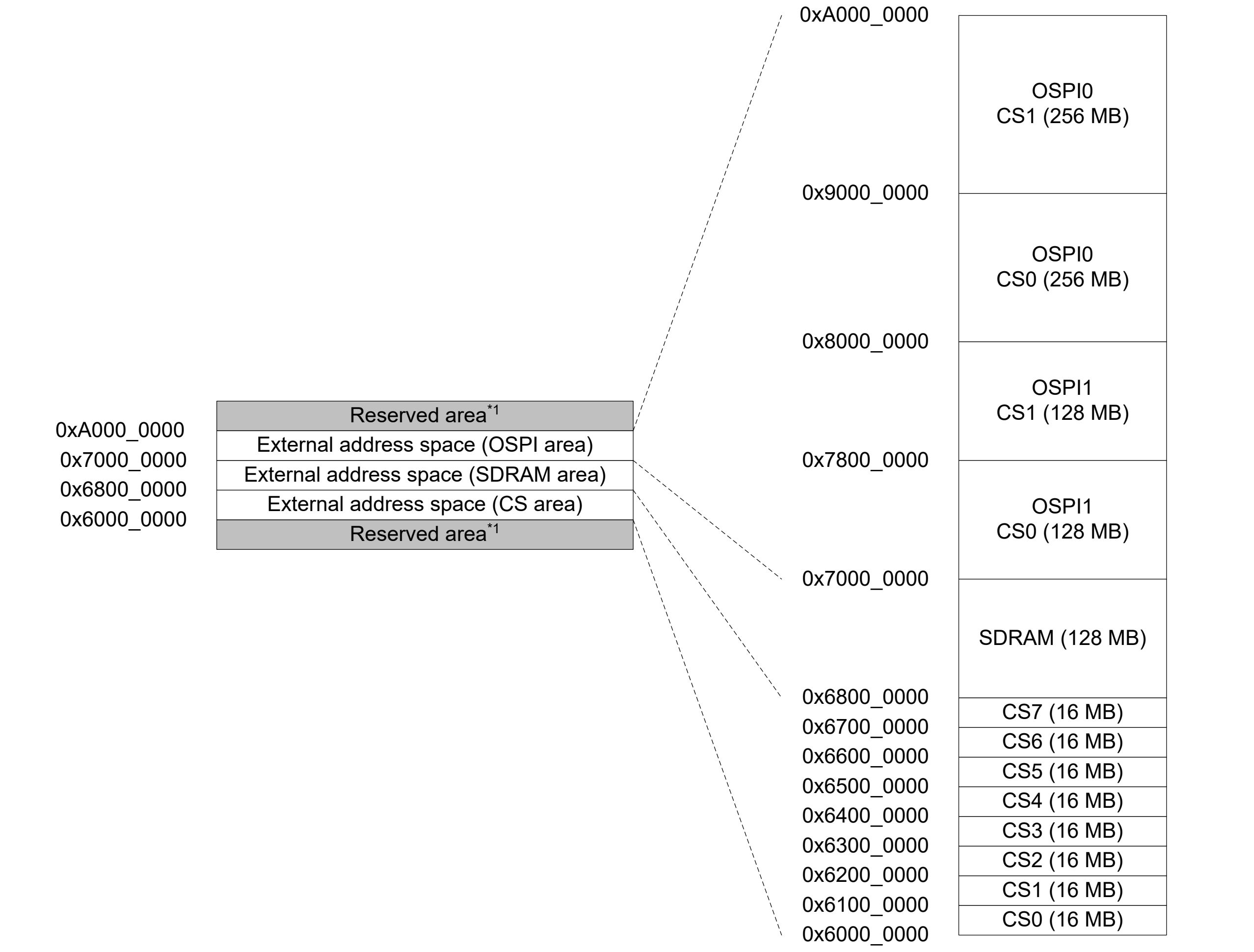
The external address space is divided into CS areas (CS0 to CS7), SDRAM area (SDCS), and OSPI areas (CS0 and CS1). The eight CS areas (CS0 to CS7) each correspond to the CSn signal output from a CSn (n = 0 to 7) pin. The two OSPI areas (CS0 and CS1) each correspond to the OM_CSn signal output from an OM_CSn (n = 0, 1) pin.

Figure 5.3 and Figure 5.4 show the address ranges associated with the individual CS areas (CS0 to CS7), SDRAM area (SDCS), and OSPI areas (CS0 and CS1).



Note 1. Reserved areas should not be accessed.

Figure 5.3 Detailed address map for CS area, SDRAM area, and OSPI area for SiP Product



Note 1. Reserved areas should not be accessed.

Figure 5.4 Detailed address map for CS area, SDRAM area, and OSPI area for Standard Product

5.3 Peripheral I/O Register Address Space

The peripheral I/O register address space is classified into two types. Figure 5.5 shows the address space types and Table 5.2 shows the address space types for each module.

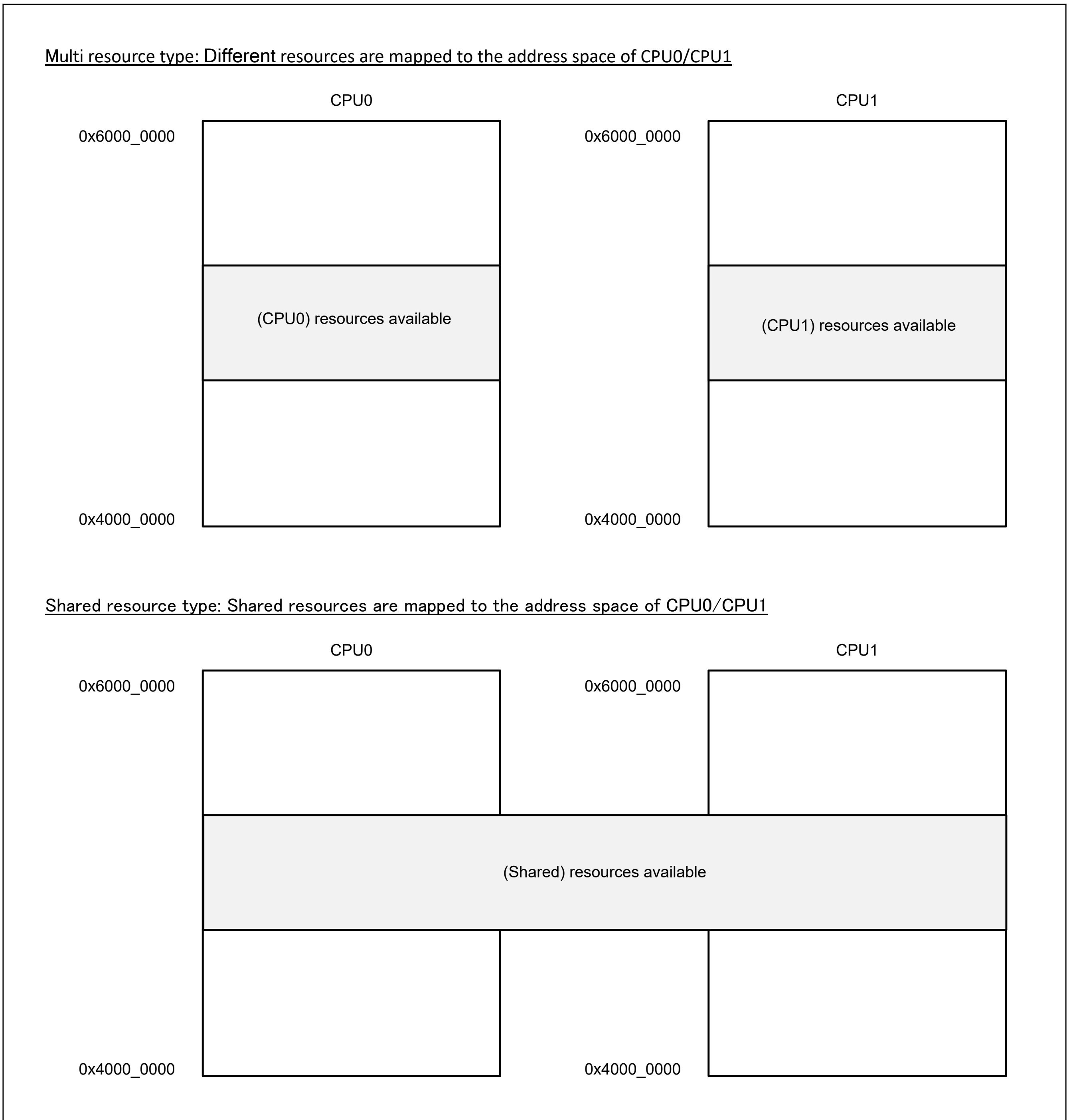


Figure 5.5 Address space types

Table 5.2 Address space types of each module (1 of 2)

Module name	Register name	Space type
ICU	NMIER, NMICLR, NMISR, WUPEN0, WUPEN1, DSLPWUPIRQENj, DELSRm, IELSRn	Multi resource type
	Other than the above	Shared resource types
DMAC	DMACCHSAR, DMACCHPAR, DMACSAR	Shared resource types
	Other than the above	Multi resource type
DTC	DTCSAR	Shared resource types
	Other than the above	Multi resource type

Table 5.2 Address space types of each module (2 of 2)

Module name	Register name	Space type
Others	All	Shared resource types